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A pipeline for harmonising NHS Scotland laboratory data to enable national-level analyses

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Abstract

Objective

Medical laboratory data together with prescribing and hospitalisation records are three of the most used electronic health records (EHRs) for data-driven health research. In Scotland, hospitalisation, prescribing and the death register data are available nationally whereas laboratory data is captured, stored and reported from local health board systems with significant heterogeneity. For researchers or other users of this regionally curated data, working on laboratory datasets across regional cohorts requires effort and

time. As part of this study, the Scottish Safe Haven Network have developed an open-source software pipeline to generate a harmonised laboratory dataset.

Methods

We obtained sample laboratory data from the four regional Safe Havens in Scotland covering people within the SHARE consented cohort. We compared the variables collected by each regional Safe Haven and mapped these to 11 FHIR and 2 Scottish-specific standardised terms (i.e., one to indicate the regional health board and a second to describe the source clinical code description)

Results

We compared the laboratory data and found that 180 test codes covered 98.7% of test records performed across Scotland. Focusing on the 180 test codes, we developed a set of transformations to convert test results captured in different units to the same unit. We included both Read Codes and SNOMED CT to encode the tests within the pipeline.

Conclusion

We validated our harmonisation pipeline by comparing the results across the different regional datasets. The pipeline can be reused by researchers and/or Safe Havens to generate clean, harmonised laboratory data at a national level with minimal effort.

Keywords: medical laboratory data, harmonisation, open-source software pipeline

1. Introduction

Using routinely collected healthcare data for research, also called secondary use of healthcare data, is important for answering population-level health questions to inform public health decision making [1–5]. Medical laboratories perform tests on samples of blood, urine, or other substances from the body [6]. Linked to other healthcare information such as demographic (e.g., age, biological sex, socioeconomic factors), medications (e.g., antibiotics, antiseptics) and medical conditions (e.g., Alzheimer's, Parkinson's) laboratory data provides a powerful basis for data-driven medical research [7].

Health laboratory data is an important part of Electronic Health Records (EHRs). EHRs are sensitive data [8], and their sharing, storage and use are regulated by national and

international laws [9,10]. The Data Safe Haven provides a platform for the use of NHS electronic data (EHRs) in research feasibility, delivery and pharmacovigilance. Working to agreed principles and standards these Safe Havens provide access to health data and services to enable research while protecting the confidentiality of the data. Data remains under the control of the data controller and complies with legislative and data controller's policies. In Scotland, health data access is provided through the Scottish Safe Haven network (SSHN) comprising four regional Safe Havens (DaSH [11], DataLoch [12], Glasgow [13] and HIC [14]) and one national Safe Haven known as electronic Data Research and Innovation Service (eDRIS)) [10,15]. The eDRIS manages all the nationally collected and curated healthcare datasets (e.g., Scottish Morbidity Records, Hospital Outpatient, Prescribing data etc.). The regional Safe Havens work closely with their local Research and Development nodes of the National Health Service (NHS), accessing regionally collected NHS data (e.g., microbiology data, virology data, laboratory tests, and deeply phenotypic clinical datasets like stroke and echocardiology data) in addition to a subset of the nationally collected datasets pertaining to the records from the population within their region. The SSHN covers around 95% of the Scottish population [16]. In Scotland, NHS laboratory data are not collected and curated at a national level and laboratory information management systems (LIMS) are managed by the individual health boards. To support researcher access to data the regional Safe Havens work with the local health boards to extract data from the LIMS and make them available for regional-level studies. Bringing data together from multiple regions or health boards is challenging due to LIMS in different laboratory has different data structure and processes.

Data harmonisation transforms multiple datasets into a single unified format/standard, which makes them more comparable [17]. There are a variety of methods of harmonisation EHRs data, e.g. Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) [18,19], i2b2 CDM [20], Fast Healthcare Interoperability Resources (FHIR) standard [21], Virtual Data Warehouse (VDW) CDM [22] and Food and Drug Administration Sentinel CDM [23]. Various CDMs have various data structure standards and vocabulary dictionaries (terminology). Adopting a CDM improves the data comparability, as it standardises the data structure as well as vocabularies. However, data in the same CDM are not always guaranteed to be comparable, for instance, same laboratory test results can be recorded under different test unit and the quantity value as quantitative measures may have collected differently and need to be harmonised

before they can be compared. There is yet not one CDM that is widely adopted across the globe. In this paper, to increase reproducibility and reusability we adopted the FHIR standard [24]. FHIR standard is a set of rules and specifications for exchanging electronic health care data. It is designed to be flexible and adaptable so that it can be used in a wide range of settings and with different health care information systems [21]. In this study, we concentrate on a simpler, rapid and flexible approach to harmonising laboratory data based on FHIR specification. The broader adoption of OMOP for the full spectrum of EHR in the contest of COVID-19, including laboratory test data, is detailed in a separate report [25].

Laboratory data is highly heterogeneous [26], thus significant efforts are needed to harmonise data from different sources prior to secondary usage. EHR data, including laboratory data, are collected for patient care and not designed to support research across multiple laboratories or regions. The different laboratories may have made different decisions around test assay kits [27], operating procedures and reporting formats [28]. The variety of data available means that comparisons are difficult, and the data needs to be harmonised prior to any analyses being possible. Currently, researchers need to this repeatedly themselves with varying levels of knowledge. We have an opportunity in Scotland to do this harmonisation process once and for all using the expertise of all the Safe Havens.

Tate et al [29] defined harmonisation of laboratory testing by referring to the same interpretation irrespective of the measurement procedure used, the unit or reference interval applied, and when and/or where a measurement is made. Much of the work in this field is to promote harmonisation on the testing process [30]. Flatman [31] systematically discussed the significance of laboratory data harmonisation as well as its potential impact. His paper recognised the future directions in reporting involves the need for a strong harmonised foundation. Pooling health related laboratory data is difficult due to the data governance, deidentification and linkage required [1]. A limited number of research studies are reporting the harmonisation methodologies of health laboratory data. Nam et al [32] compared five clinical chemistry instruments from 2015 to 2019 for 12 clinical chemistry measurements using pooled residual patient samples for weekly verifications. An automated data cleaning tool was developed for Scottish laboratory data curation which automatically detects and replaces wrongly spelled units, and performs a unit conversion as well as outlier detection [33]. A protocol for cleaning

paediatric height and weight data incorporating outlier detection has also been developed and optimised [34]. In Scotland, previous work on laboratory data mainly focused on the mapping of variables/coding/vocabulary where biochemistry data comprising 145 tests were LOINC encoded for anonymous data obtained from HIC [35] [36]. To the best of our knowledge there are no other studies in the literature performing multi-regional health laboratory data harmonisation in a national level scale covering wide spectrum of tests.

Problem or Issue	For researchers or other users of this regionally curated data, working on laboratory datasets across regional cohorts requires effort and time.
What is Already Known	Laboratory data are separately hosted in four regional Safe Havens with high heterogeneity
What this Paper Adds	We developed an open-source software pipeline to create harmonised laboratory datasets for researchers

Table 1 Statement of significance

In this paper, we are reporting the harmonising of cross-region sensitive health laboratory data from all four regional Safe Havens in Scotland. The aim was to create tools for researchers to work with whole-of-Scotland population laboratory datasets. The objectives were 1) to obtain data from each regional safe havens, 2) to identify the most common assays/results, 3) to generate comparative analysis between sources, 4) to share tools for harmonising the data. For the first time, we successfully pooled a laboratory dataset that covered all major Scottish health boards. Working closely with local Safe Havens and laboratories, cooperating the local knowledge on data, we mapped heterogeneous data into FHIR standards and harmonised the contents of the data to make them comparable. This work will aid researchers reuse this data and improve reproducibility in complex and heterogeneous dataset of significant utility in healthcare research for national-level analyses which are currently challenging to achieve in a timely manner.

Kahn et al. [37] propose two equally valid approaches for Data Quality (DQ) control: Verification and Validation. Verification examines how well the data align with expectations based on metadata constraints, system assumptions, and local knowledge. In contrast, validation involves comparisons against known truth, relative gold standards,

or external resources independent of the data source. They further categorised data quality assessment into three key dimensions: (1) Conformance, (2) Completeness, and (3) Plausibility. This paper focuses entirely on data quality verification. The Materials and Methods section addresses Conformance, Completeness is discussed in Results sections 3.1 to 3.3, and Plausibility is evaluated in the technical verification results presented in sections 3.4 to 3.5.

Harmonisation of laboratory data contains a wide spectrum of topics, e.g. test unit, quantitative test results, none-quantitative test result, test results censorship, etc. Extract, Transfer and Load (ETL) is the process of combining data from multiple sources into a large, central repository. The ETL process, especially the “transfer” step are local data specific. It is not possible to cover it all in detail. In this paper we are using our knowledge of the regional Scottish Safe Haven Network to showcase an open-source pipeline and process to reproducibly combine routine laboratory data for research. The principles we describe including details of the reason, meaning and effects (with examples) will provide guidance to other regions.

2. Material and Methods

The Scottish Health Research Register (SHARE) is a consented population-level dataset and biobank with over 300,000 registered participants [38]. It provides research access to routinely collected NHS data for SHARE participants and allows determination of suitability for research projects/clinical trials.

We completed the required data governance applications and extracted laboratory tests (biochemistry, haematology, immunology, virology, pathology and microbiology) from all registered SHARE participants who were residents from 01/01/2015 to 31/12/2021 (or until death) in the region of each of the four regional Safe Havens. The data were de-identified and provided as project data for analysis within the HIC Trusted Research Environment [16].

Table 2 details the data records received. Not all Safe Havens provided access to all laboratory data sources with only biochemistry and haematology provided by all. Typically, patients have many data records (one record equals to one entry in the database) at each request of test. The largest number of patients and therefore records are from Tayside and Fife which is a historical artefact due the SHARE project starting in that region and therefore larger numbers of participants overall.

Safe Haven	Regions Covered	Data Type	No. Records*	No. Patients
DaSH	Grampian	biochemistry, haematology	19,085,610	27,338
DataLoch	Lothian	biochemistry, haematology, immunology	10,046,575	21,529
Glasgow	The west of Scotland	biochemistry, haematology, immunology, virology, pathology and microbiology	14,953,293	23,724
HIC	Tayside and Fife	biochemistry, haematology, immunology, virology, pathology and microbiology	34,644,911	78,334

Table 2 Data Summary. The time coverage was 01/01/2015 – 31/12/2021 across four Safe Havens.

** one record equals to one entry in the database. In clinical practice, a medical professional normally requests a set of tests to be done in one request, which will result in several test records being added to the database.*

2.1. Data Flow

In Scotland, laboratory data is collected and managed by specific clinical laboratories using the laboratory management software APEX [3] as designated by the local NHS board. Data were shared by individual regional Safe Havens on agreed terms with the data controller (e.g. local NHS board). As shown in Figure 1, HIC and Glasgow Safe Havens' authorised staff extracted laboratory data from the Scottish Care Information (SCI) Store [39], an information repository that provides clinicians with secure access to patient information at the point of care. Glasgow obtained all laboratory data from SCI Store whereas HIC obtained four sub-types of data from SCI Store, with the exception of microbiology and pathology data coming directly from the laboratories in the region. Glasgow SCI data was generated from multiple sources and systems and compiled by Glasgow Business Intelligence through an ETL. This formed the source laboratory data with views of varying complexity. Glasgow Safe Haven extracted fields from this source SCI data to generate project ready files (the one we received for this project). They did not extract a field that reports the non-numeric value of a test. HIC's data included both numeric and non-numeric data. DataLoch and DaSH Safe Havens have individual bespoke systems which they used to extract their laboratory data from APEX. Lothian's Patient Management System for Secondary Care (TRAK) provided only numeric test results for secondary care. If a patient had not attended a hospital or the result was non-numeric value (e.g. positive/negative), data was not transferred from APEX to TRAK, and therefore we did not receive this. We were provided numeric and non-numeric data

from biochemistry and haematology from the Grampian Data Platform. However, since the completion of this study Grampian can now provide immunology, virology, pathology and microbiology for research projects on a per-project basis.

This study curated and analysed laboratory data up to test dates of “31/12/2021”. Regional Safe Havens receive regular data updates of laboratory data, most updated daily while some monthly.

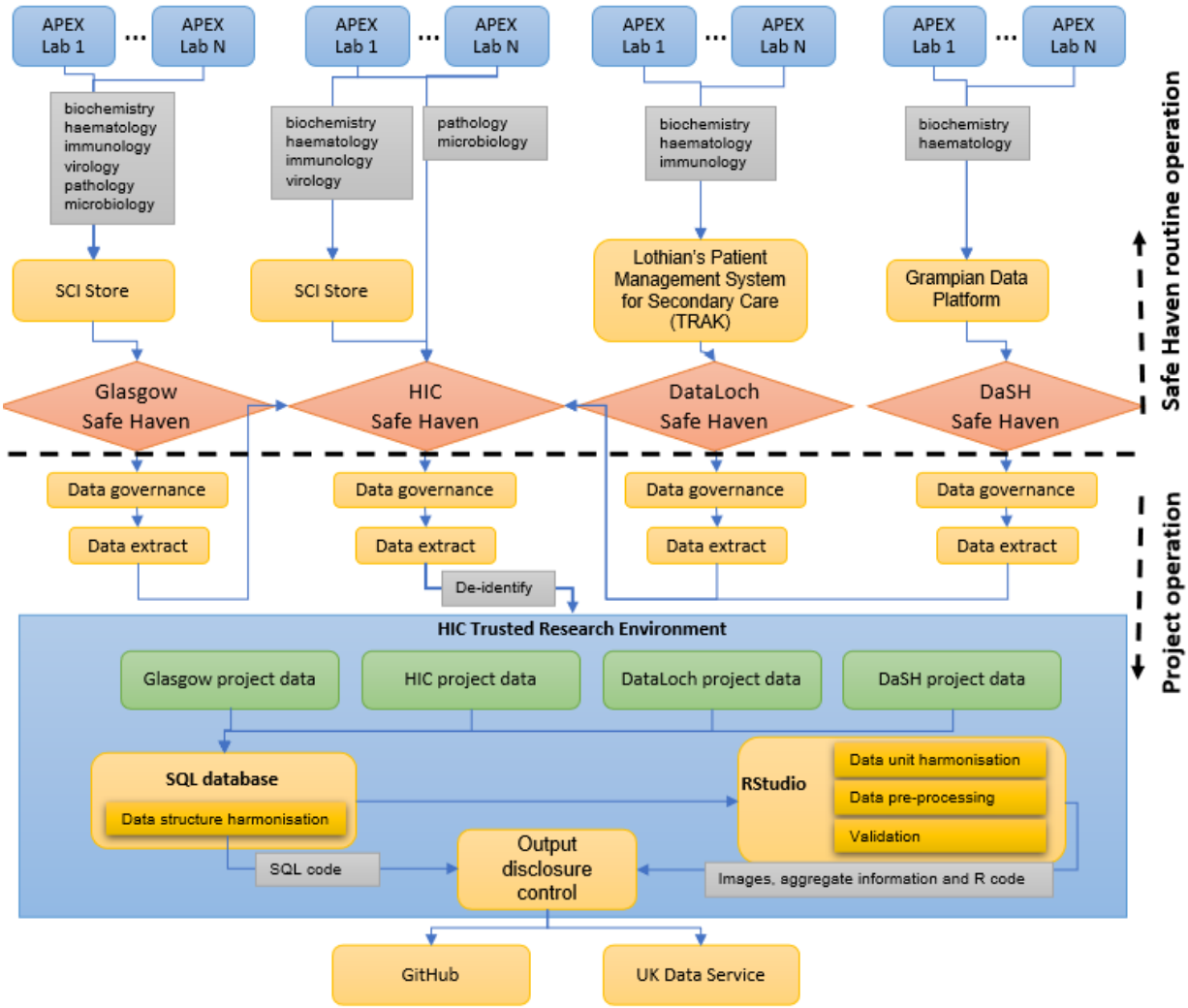

Figure 1 Data collection and processing workflow. The top half (separated by dash line) shows the routine operation of Safe Haven on collecting laboratory data. The bottom half shown the processes of data flow designed for this project.

The data was provided as flat files from the Safe Havens to HIC and loaded into a Microsoft Structured Query Language (SQL) Server database (version 16.0.1) for ease of handling the large data volumes. Dataset profiling was conducted using WhiteRabbit, a tool that generates metadata of the datasets providing descriptions of tables, fields and values with each field [40]. Data exploration and harmonisation were conducted

using a mixture of SQL and R (version 4.1.3) scripts [41], available via: https://github.com/HicResearch/ScottishLabData.git. The steps in developing the harmonisation pipeline were as follows:

Following governance approvals (individual Caldicott [10] with each Safe Haven and SHARE committee) data were transferred into HIC via the established secure route as CSV formatted flat files. Flat files from each Safe Haven were scanned using the dataset profiling tool WhiteRabbit [42] which generated a multi-tab spreadsheet “ScanReport”. The “Field Overview” tab and a “Table Overview” tab contained field and value overviews. Results of “Field Overview” on flat files received are presented in Appendix A. We cannot include the “Table Overview” due to the disclosure control e.g. patient ID being listed in the file, the frequency count lower than 5 for some row level information.

WhiteRabbit results were imported into Rabbit In a Hat [43] which provides a graphical user interface to allow a user to connect source data to tables and columns within the target data structure. The mapping established was confirmed through meetings with local data experts and/or laboratory staff. The final mapping used is shown in Figure 2. Section 3.2 describes the mapping of source data field to FHIR terms in more detail.

Based on the mapping established, SQL codes were developed to conduct the extract, transform, and load (ETL). We imported the original data into SQL database through flat files. We then extract information from the original data table, transfer them into the format required by FHIR standard as shown in Figure 2 one for each Safe Haven. The transformation SQL code created were named as “<SafeHavenName>2FHIRscot.sql” and are freely available via <DOI 10.5281/zenodo.13310959 >.

All downstream analysis of the transformed data was performed with R such as overlapping analysis (reported in Figure 3), Unit harmonisation (reported in section 2.5), visualisation and statistical verification (reported in section 3.5) and are also freely available via <DOI 10.5281/zenodo.13310959 >.

To maintain the reproducibility and transparency. We have chosen to conduct our pipeline developments with open-source software, WhiteRabbit, Rabbit In a Hat and R. We have provided the WhiteRabbit scan reports, the SQL code for ETL and R code for

harmonisation and visualisation as Appendix of the paper. All Appendix and code files were made available via MIT license and hosted in GitHub repository [44].

2.2. A common structure

HIC's laboratory data came in different structures with each subtype in one flat file, while other Safe Haven's data came in one structure. Glasgow laboratory data were provided in one flat file. DaSH's biochemistry and haematology data were provided in two separate flat files of the same structure. Lothian's laboratory data were provided based on the year of the test with each year's data in a separate file with the same structure.

Based on best practice of "Reproducible Analytical Pipelines" [4], we focused on selecting variables for laboratory data that would support the most research and promote sharing of laboratory data across regions [45,46]. As shown in Figure 2, 13 columns (including 11 representative FHIR concepts [21,47,48] as well as two other bespoke Scottish concepts, 'healthBoard' and 'readCodeDescription', as highlighted by orange arrows in Figure 2) were considered as the core set of variables. FHIR [21] is the global industry standard for passing healthcare data between systems. It is free, open, and designed to be quick to learn and implement.

Using WhiteRabbit, sampling one million rows in each flat file, we tested the data types, completeness (fraction empty) and consistency (N unique and fraction unique) of each Safe Haven's laboratory results. WhiteRabbit scan reports are provided in Appendix A. Together with data managers from each Safe Haven, we established the mapping relationship between the source laboratory data to the FHIR [49] based same structure in Figure 2. We used the Rabbit-In-a-Hat [40] software to read and display WhiteRabbit scan results. Rabbit-In-a-Hat allows a user to connect source data to tables and columns within the target data structure through a graphical user interface. Based on SQL skeleton auto-generated by Rabbit-In-a-Hat, further editing of the mapping was made based on consulting local Safe Havens to establish the final SQL script for structural harmonisation.

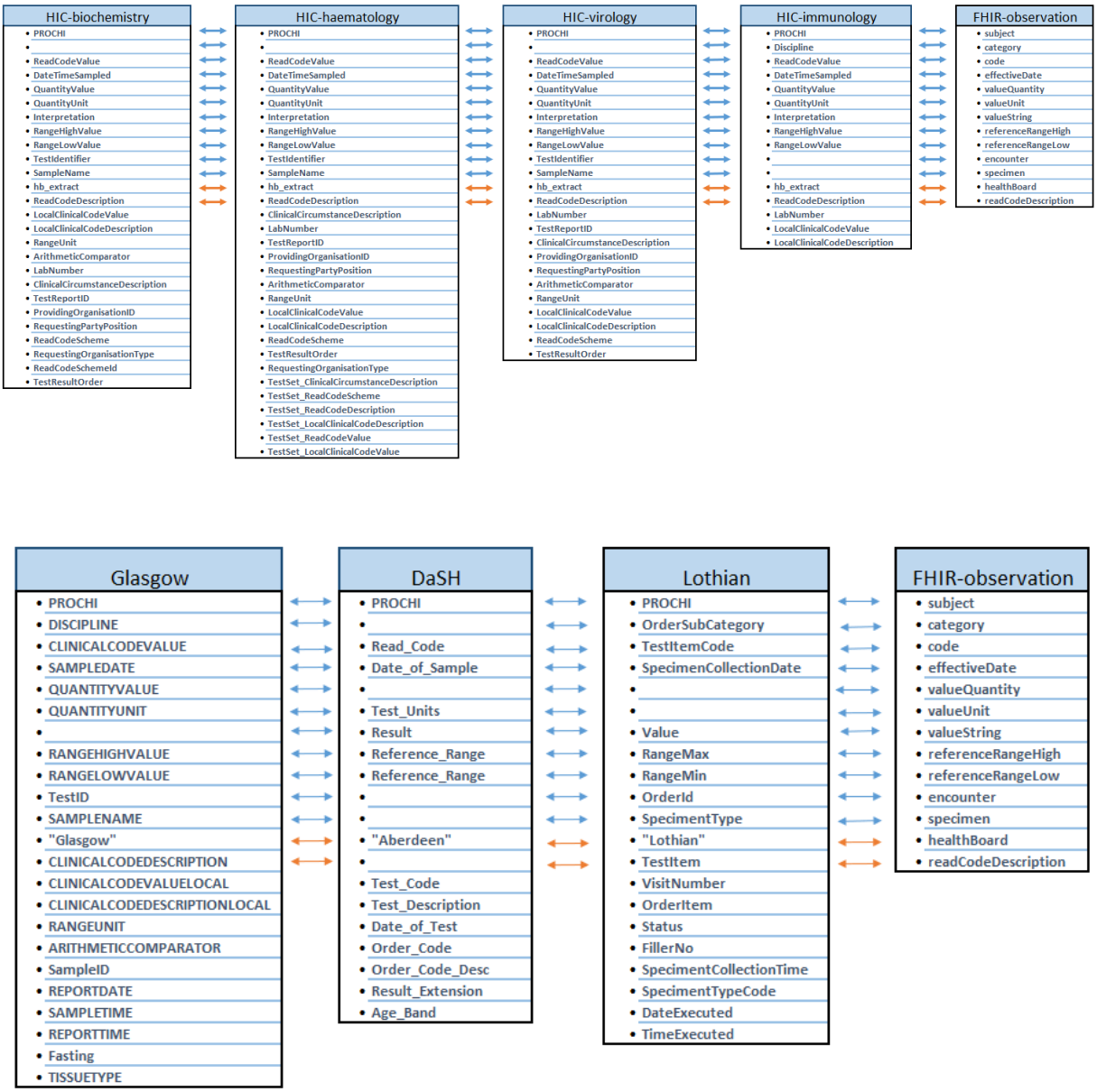

Figure 2 Selection of the 13 core variables. 11 of them (highlighted in blue arrow) are from the FHIR observation table v5.0.0 [48], and two (highlighted in orange arrow) are bespoke to Scottish laboratory data. The top half of the figure is the mapping relationship of tables for HIC data while the bottom half is mapping Glasgow, DaSH and Lothian data into the same structure.

Standardising censored result

Laboratory test results may be left-censored, known to be less than some lower limit (measurements below a limit of detection) or right-censored, which contains measurements above the maximum value that can be detected [50,51]. In FHIR standard structure, test results were recorded in “valueQuantity” and “valueString” columns. “valueQuantity” for numeric value and “value String” for none-numeric results e.g. ‘positive’/’negative’. Our pipeline put censored results (‘<LowThresholdValue or

‘>HighThresholdValue’) in the ‘valueString’ concept, meantime the numeric value of the censored results in the ‘valueQuantity’ concept.

All Glasgow test records were provided with the quantity value only. For Glasgow Safe Haven’s censored test results, “valueString”(s) were reconstructed by combining the “ARITHMETICCOMPARATOR” and “QUANTITYVALUE” concepts in the original data. For example, one of eGFR test (451E.) returned “60” in “QUANTITYVALUE” and “>” in “ARITHMETICCOMPARATOR” in the original Glasgow dataset. This test result was recorded as “60” in “valueQuantity” while “>60” in “valueString” in our standard structure. DataLoch and DaSH recorded all censored test results as “<LowThresholdValue” or “>HighThresholdValue”. For censored test results of those two Safe Havens, “valueQuantity” concept was reconstructed by extracting the numeric value from the censored results. Taking one of eGFR test results in DaSH data as example, the result was originally recorded in “Result” column as “>60”. With our pipeline, the numeric value “60” was extracted and put in “valueQuantity”, while “>60” was used as “valueString”. In HIC’s original data structure, some censored results had “<LowThresholdValue” or “>HighThresholdValue” recorded in the “Interpretation” column while “ArithmeticComparator” was empty, the rest had “<” or “>” in the “ArithmeticComparator” column while “Interpretation” column was empty. Our pipeline reconstructed both “valueQuantity” and “valueString” for HIC’s censored records based on the entries in “Interpretation” and “ArithmeticComparator” in the original data.

2.3. Coding standard

The coding standard in Scottish laboratory data is based on Read Code. Read Codes [34] as a coded thesaurus of clinical terms has been used in the NHS since 1985. There are currently two versions: version 2 (v2) and version 3 (CTV3 or v3). The majority of the data extracts we received were captured in Read Code Version 2 [34], however, one source used local test codes as their primary reference. We needed to map the local codes to read codes in collaboration with the laboratory resulting in 71M out of 78M (91%) records received being either directly annotated with Read Codes or had a mapping file linking local testing code to Read Code. We selected records with Read Codes for further analysis. Read v2 and v3 (CTV3) are being retired [52]. NHS England has established a mapping of Read Code to SNOMED CT [53] which can be applied to Scottish data. We have provided the Read Code to SNOMED CT code mapping of the 180 tests (selecting method described below) in Appendix C.

2.4. 180 selected tests

There were 2138 unique Read Codes (HIC 875, Glasgow 1438, DataLoch 587 and DaSH 407) across all four Safe Havens. The difference in codes is due to both variations in local practice (we observed many self-developed read codes in Glasgow) and variation in laboratory test types (Glasgow and HIC have the full collection while DaSH and DataLoch only had two or three types of laboratory tests). Focusing on those with the highest coverage (record count), the top 100 recorded Read Codes of each Safe Haven were selected resulting in a total of 180 unique Read Codes. These 180 codes cover 98.7% of all laboratory tests records across the country (HIC 98.9%, Glasgow 97.4%, DataLoch 98.5% and DaSH 99.5%). All percentages were calculated after excluding those tests without Read Codes. As shown in Figure 3, there was a large overlap between Safe Havens and the Read Codes used with 87 (covering 79.1% of records) present in all Safe Havens and 111 (covering 87.6% of records) in 3/4 Safe Havens. Appendix B provided a lookup table of the 180 test Read Codes and test descriptions.

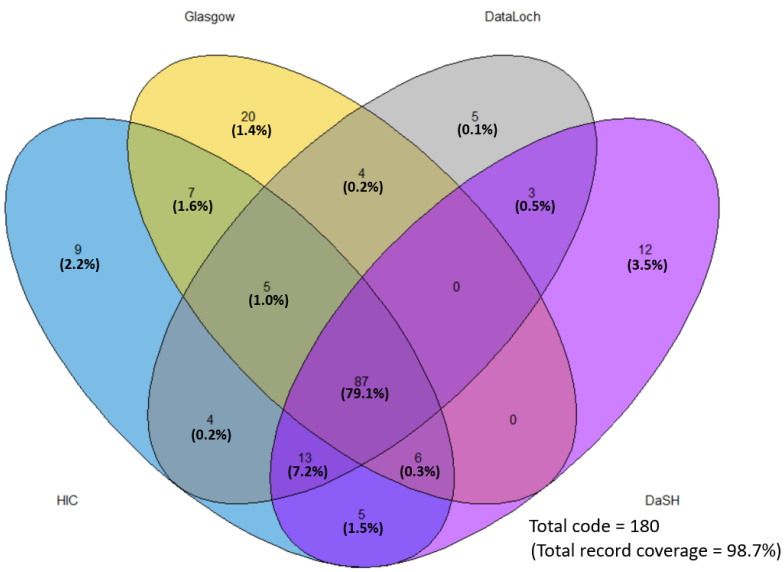

Figure 3 The overlap between the 180 selected Read Codes across DaSH, DataLoch, Glasgow and HIC Safe Haven, with code count in each set in black and total records coverage percentage in bold black

Some of the 180 codes are not normally considered as laboratory tests. We included those codes in our pipeline as they are informative of the local laboratory and data collection practise. For example 'haemolysis' (42h6.) and 'lipaemia' (4QA..) both had around 17 counts per thousand patients in DaSH data while there were no instances of

this test carried out in all other regions. There's a third test in this group called ‘icteric’ (44p..). Those codes referring to the condition of the blood specimen received. Each of these conditions will have an impact on whether a test can be analysed or whether a repeat blood draw is required. In DaSH laboratories those have been implemented as test codes for ease of data entry and for stats reporting. For haemolysed, lipaemic or icteric specimens, an increasing number of "+" symbols (between 1 and 4) was entered as a result for the relevant test, so a strongly lipaemic sample would have a LIP result of ++++.

A similar practise was seen in DataLoch’s data which used ‘haemolytic complement’ (43I5. but not included in the 180 codes due to low records counts) to indicate the sample is fit for analysis. In HIC data, ‘SAMPLE LIPAERMIC’ or ‘SAMPLE HAEMOLYSED’ was recorded in ‘valueString’ to disqualify the specific test. We did not observe recording of any of the above for sample quality in Glasgow data. Likely, the information was not extracted from the source SCI data.

2.5. Unit heterogeneity

Two types of unit heterogeneity were observed. Firstly, tests from different laboratories had different units for the same test due to the variation of test methods. For example, most of the Glasgow tests for creatinine urine (46M7.), were in ‘mmol/L’ with 4 of the tests in ‘µmol/L’. In contrast, the majority of DaSH tests were in ‘µmol/L’ with 25 of the tests in ‘mmol/L’. All HIC and DataLoch tests used ‘mmol/L’ as the unit of measurement. Secondly, lack of consistency in data collection were observed. For example, in the carboxyhaemoglobin blood test (42Y0.), 19 data entries out of 29819 entries in Glasgow data had ‘% Total Hb’ in their ‘valueUnit’ concept while in the rest of the data, including data from other Safe Havens, ‘%’ was used in ‘valueUnit’ concept. The variation between ‘% Total Hb’ and ‘%’ shown the lack of consistency across health boards for data collection. For the urine protein to creatinine ratio test (44lD.), all Glasgow tests were in ‘mg/mmol creatinine’, all DaSH tests were in ‘mg/mmolcr’ while all HIC tests were in ‘mg/mmol’. For the 87 Read Codes common across Safe Havens (see Figure 2) 34 of them used the same test unit across all regions.

For each of the 180 Read Codes, the most frequently used units were defined as the standard test units. Test results were changed to match the standard units. Transformation of the “valueQuantity” and “valueUnit” columns was conducted

accordingly to change all test records to the same standard unit. Appendix B shows the standard test unit and the transformation relationship used. For example, as shown in Appendix B, for potassium test (44I4.), HIC records used both mmol/L and nmol/L while Glasgow, DaSH and DataLoch records use mmol/L. The "if" column in Appendix B is "if nmol/L value 1e-06" which is interpreted as per the following pseudocode.

```
If "valueUnit" equals to "nmol/L" {
        Multiply "valueQuantity" with 10^-6, using the results as new "valueQuantity"
        AND
        Change "valueUnit" to "mmol/L"
}
```

For records with a non-standard unit that could not be transferred to standard (1.0% of total records), we excluded those tests from further analysis. For example, haemoglobin (423..), there were "nonsense unit" such as a name or some other string, i.e. 'HARRY' have been excluded from further analysis.

2.6. Technical verification

We worked closely with local data experts and laboratories to maintain a high standard in mapping and harmonisation pipeline development. There was still a need to assess the accuracy and effectiveness of our pipeline. Due to the diverse test types (180 codes) and population covered, it was difficult to select one validation metric that works for all. We used comparison methods including high level statistics mean (standard derivation), median [Interquartile range] (see Table 4), visualisation such as density distributions, boxplots and histogram plots to validate our result dataset (see Figures 4,5 & 6).

For technical verification only, we applied pre-processing data cleaning as described below to all data before conducting data verification and consistency checks for significant discrepancy across Safe Havens. The pre-processing methods we adopted were for data verification. It is not part of the pipeline for delivery of data. Individual studies should choose their pre-processing methods according to their specific requirements.

Our data do not meet the assumptions for a parametric test; therefore, we chose a non-parametric alternative. The Kruskal-Wallis test is a widely used statistical method to test the null hypothesis that the medians of three or more groups are equal.

Replacing censored results with a specific value (such as limit of detection (LOD)/2 or LOD/$\sqrt{2}$ for left censored value) is a common practise [50,51]. For sense checking, we replaced all left-censored values with LOD/2 and right-censored values with LOD×2. The approach changed 5.8% of the total records.

For data visualisation, it was necessary to apply an outlier cut to exclude extreme values from the distribution plot to focus on an appropriate range. Van den Broeck et al. [54] specified the true normal, true extreme, erroneous and idiopathic values for each test code and discussed the outliers case by case. These methods cannot be easily applied to big datasets with many test codes. Typical outlier detection (lower outer fence: Q1 (first quartile)-3xIQR (Interquartile range), upper outer fence Q3 (third quartile)+3xIQR) could not be safely applied because "true" results from diseased patients can be far from healthy ranges. Some tests (liver function tests) had values which are far from the healthy range but still valid. We used a lower outer fence: Q1-5xIQR, upper outer fence Q3+5xIQR to filter extreme numbers and changed the result to "NA".

We visualised the data for verification and sense checking. Density distributions and boxplots were used for both raw and pre-processed data for each test to understand the impact of pre-processing on the data. We included histogram plots for each Safe Haven showing the test value distribution in that Safe Haven based on pre-processed data. We then combined the patient demographic information with the multi–Safe-Haven laboratory test results to show the value variation in each demographic group for each of the selected tests.

3. Results

3.1. Cohort Demography

Demographic data were linked to all the laboratory data. Age was calculated based on the difference between 1st of April 2023 and the date of birth of the patients. Similar age and sex distributions were observed between cohorts from different regional Safe Havens, giving reassurance that any differences are unlikely to be due to population differences. We also compared indices of deprivation of SHARE cohorts from each Safe Haven and confirmed that the SHARE cohort has a good representation of the national demographic as shown in the Scottish Index of Multiple Deprivation 2020 [55].

3.2. A national level laboratory data

The percentage of completeness for each FHIR concept after transformation into a common data structure is shown in Table 3 for each individual Safe Haven and across all of them. DaSH does not capture “encounter” or “specimen” and so those fields were recorded as “NA". Column “subject” contained a de-identified version of CHI. Individuals in Scotland are assigned a Community Health Index (CHI) number [56] when they first interact with the health service which is retained within their EHR throughout their health record history. Safe Havens use CHIs to link datasets. Column “category” contained one of the following six values: biochemistry, haematology, immunology, virology, pathology and microbiology, representing the laboratory data subtypes. Column “code” was Read Code for that test. Details are provided in the coding standard section. Column “effectiveDate” was the date the test was conducted. “valueQuantity”, “valueUnit” and “valueString” contained the test results. Details of how “valueQuantity” and “valueString” differ from each other are provided in a common structure section. “encounter” was the test or order ID. “specimen” described the type of specimen used for the test. “healthBoard” indicated which health board the test was conducted. Safe Haven name was used for “healthBoard” if information was missing from original data. “readCodeDescription” was the description of the test. For the same Read Code, multiple descriptions were observed in both inter- and intra-Safe Haven data. The descriptions all had the same meaning but a slight difference in the wording. We retained the original test description as they help to validate the read code.

Field	Type	DaSH	DataLoch	Glasgow SH	HIC	ALL
subject	String value	100.0%	100.0%	100.0%	100.0%	100.0%
category	String value	100.0%	100.0%	100.0%	100.0%	100.0%
code	String value	100.0%	100.0%	100.0%	100.0%	100.0%
effectiveDate	String value	100.0%	100.0%	100.0%	100.0%	100.0%
valueQuantity	Real number	98.4%	97.6%	100.0%	100.0%	99.00%
valueUnit	String value	73.5%	94.4%	94.6%	84.8%	86.80%
valueString	String value	7.4%	100.0%	3.2%	19.6%	32.50%
referenceRangeHigh	Real number	71.8%	88.4%	100.0%	100.0%	90.00%
referenceRangeLow	Real number	71.8%	88.4%	100.0%	100.0%	90.00%
encounter	String value	NA	100.0%	100.0%	100.0%	100.0%
specimen	String value	NA	94.1%	100.0%	100.0%	98.00%
healthBoard	String value	100.0%	100.0%	100.0%	100.0%	100.0%
readCodeDescription	String value	100.0%	100.0%	100.0%	100.0%	100.0%

Table 3 Percentage completeness (percentage of records with values) after mapping the laboratory data into the FHIR common data model. Taking Glasgow data 'valueString' concept as example, there are 3.2% of Glasgow records which have a value in 'valueString' column, meaning 96.8% of Glasgow records don't have value in 'valueString'. This is because the none-numeric value is not recorded for all Glasgow records in the original data. Those 3.2% are reconstructed and filled by our SQL pipeline for censored test results.

3.3. Test frequency

The ratio between the maximum record count and the minimum record count per 1000 patients was used to compare the differences. There were 38 tests that had more than 10-fold variation between the highest and the lowest Safe Haven.

Chloride test (44I6.) showed a 474-fold variation between DataLoch and Glasgow Safe Haven. Glasgow and DaSH had a frequency of 16 to 19 per thousand patients while HIC and DataLoch's data had a frequency of 0.04 to 0.07 per thousand patients.

The variation in frequency reflected the different test profiles offered by the different laboratories and differences in requesting by individual hospitals [57,58]. The variations may also be caused by different populations/case mix, deprivation, disease incidence/prevalence, local policy decisions, and availability of relatively new diagnostic services in speciality areas [59,60].

3.4. Reference Range

"referenceRangeHigh" and "referenceRangeLow" were the reference range of the test for the patient from the laboratory. A test result within the range was considered "normal". The reference range for a test varied both inter- and intra- Safe Havens. As might be expected, reference ranges can vary based on the demography of the patient [61]. For example, the range of results for the prostate-specific antigen test (43Z2., chosen to exclude the influence of biological sex) is shown in Figure 4. The reference range changes with the age of the patient; the higher the patient's age, the wider the reference range. Grampian data also had a 0-2.5 reference range, which is not seen in other Safe Haven test results.

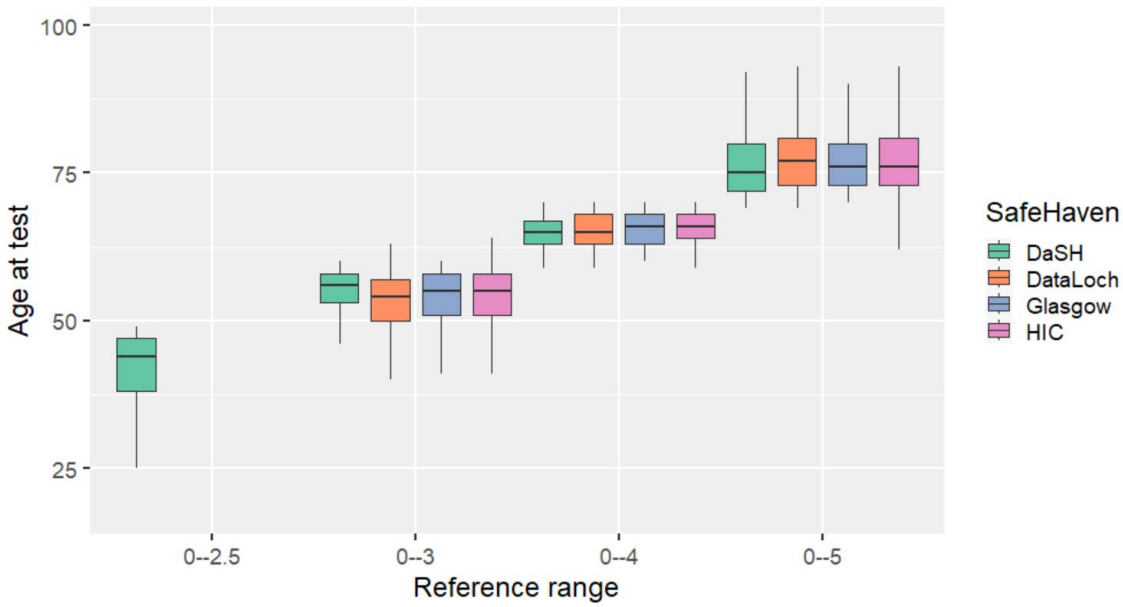

Figure 4 *Prostate-specific antigen test (43Z2.) reference range changes based on the patient's age. x-axis is the reported reference range came with the test; y-axis is the patient age at test. Each boxplot represented the age distribution for one reference range in one Safe Haven. Different colours represent data from different Safe Havens. Age is calculated as age by test (the difference between date of birth and data of test).*

3.5. Example case using serum creatinine

Given that the source data is sensitive and has single-person resolution it is not possible to report all of the results from the 180 tests. The verification for the creatinine test (44J3.) is shown below. All results in the dataset have gone through disclosure control and have minimum risk of re-identification.

Table 4 shows the general information about the creatinine test (44J3.). The "valueQuantity" row was calculated based on pre-processed data while the 'Reference Range' and 'Test Frequency' row in Table 4 was based on the raw data. The unit heterogeneity for creatinine was due to lack of standards e.g. different assay systems being used over time. "unit" was changed to standard (umol/L) without changing the test "valueQuantity". Different reference ranges were observed across Safe Havens. Test frequency was calculated based on the ratio between the total count of test and the total cohort size of the region (count/person). The test frequency between different Safe Haven remained similar.

	DataLoch	DaSH	Glasgow	HIC

valueQuantity (median[Interquartile range])	73[28]	72[29]	71[28]	75[32]
Unit	umol/L	umol/L;umol/l	umol/L	umol/L
Reference Range (Reported in original data) *	50--98; 64--111; 60--120; 53--78; 61--97; 34--72; 40--98; 39--92; 31--67; 26--57	50--100; 45--84; 59--104; 50--120; 30--98; 44--79; 32—76; 48--81; 47--98; 40--83; 54--99; 38--76; 43--75; 38--74	44--80; 62--106; 50--100; 50--120; 44--106; 54--99; 50--77; 48--81; 47--98; 44--79; 46--70	44--80; 62--106; 50--100; 50--120; 44--106; 54--99; 50--77; 48--81; 47--98; 44--79; 46--70
Test Frequency (test count/person)	16.27	15.74	18.82	13.54

Table 4 General information about creatinine test (44J3.)

** there might be many reasons for alternative Reference Range e.g demographic, test methods, test kit etc.*

Figure 5 shows a density distribution and a box plot of the quantity value for each Safe Haven for the creatinine test. It was difficult to visualise the distributions of most of the raw data results due to the impact of the extreme values on the graph scales. After preprocessing, all four regional data shown a gamma distribution [62] of density distribution. We confirmed the density distribution with the histogram of each Safe Haven.

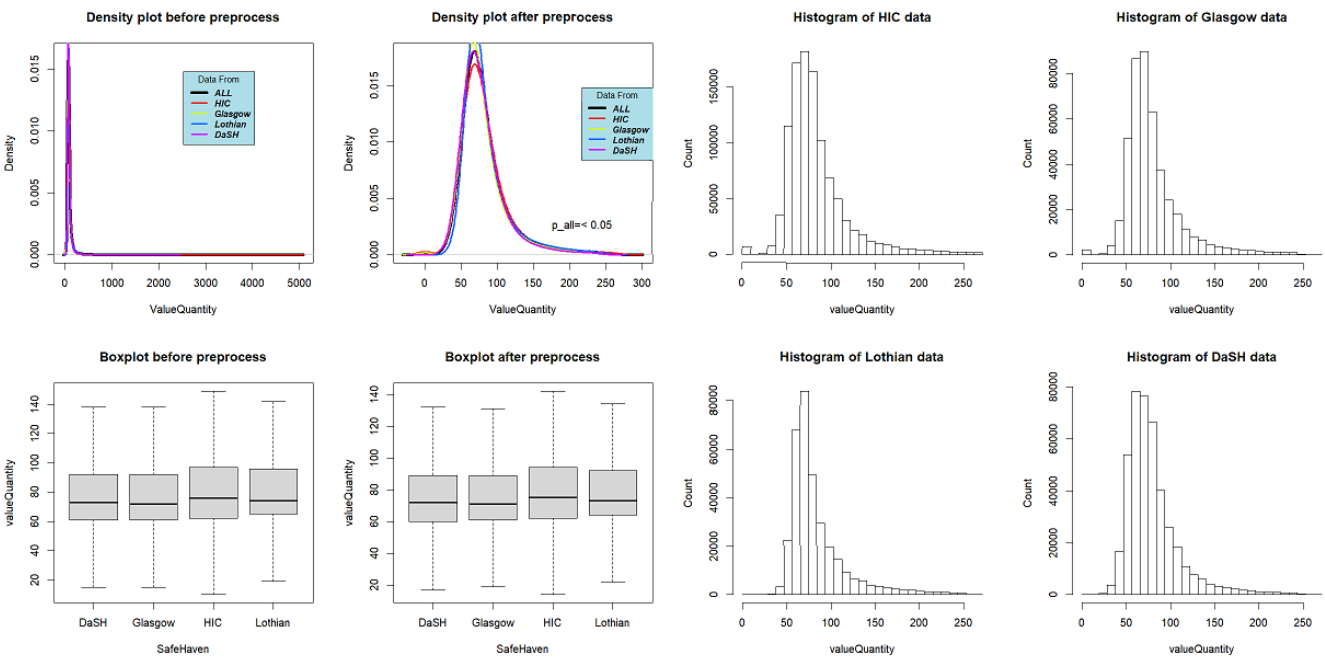

Figure 5 Quantity value density distribution and box plot for creatinine test for multi-Safe Haven dataset. The left column two figures were based on raw data while the rest figures were based on pre-processed data. The pre-processing methods were

detailed in the Technique Verification section. Density distribution plots used different colours of lines represent data from different Safe Havens. P value displayed in "density plot after preprocess" were based on Kruskal-Wallis test.

We provided the test result group by sex and age of the patient (at test) in Figure 6. The figure shows a lower test result for females (compared with male) across all age groups. The "valueQuantity" variation within the age group increased with patient age consistent with the literature [63] on creatinine test.

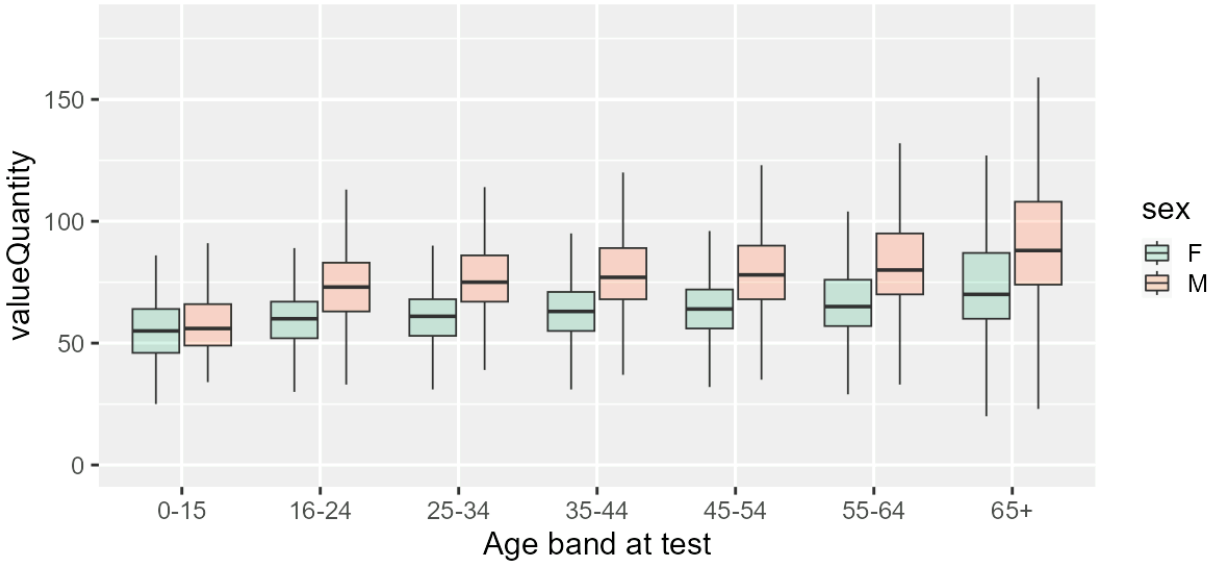

Figure 6 Creatinine test cross Scottish health board group by sex and age (at test). The x-axis is the age group while the y-axis is the test quantity value. The sex difference was represented using different colours of the box plot.

We have generated visualisations as shown in Figure 5 as well as data shown in Table 4 for all codes cover by more than one Safe Haven. For all visualisations we have calculated measures of central tendency, variability as well as value density distribution to confirm the consistency of the outputs generated by the pipeline for each code. Due to local statistical disclosure control regulations – as required by the project governance – the complete data cannot be made publicly available. Instead, we have included 10 tests' frequency and quantity Value statistic in Appendix D. There were high similarities between Safe Havens on the value central tendency and test utilisation. For example, Potassium test (44I4.'), the median and Interquartile Range for HIC, Glasgow, DataLoch and Dash were 4.3 [0.7], 4.3 [0.6], 4.3 [0.6] and 4.3 [0.5], respectively. The mean and standard deviation were 4.4 (0.5), 4.3 (0.5), 4.4 (0.5) and 4.3 (0.5) respectively.

4. Discussion

Laboratory data is very important clinically for diagnosis and patient monitoring which also feeds into the utility for observational retrospective studies. There is lack of a nation-wide harmonised source for this data. In Scotland the laboratory data is managed by the local laboratories and used for research purposes by the regional Safe Havens.

This research demonstrated a process for bringing together a national, linked dataset of 151,000 patients and 78 million laboratory records for the first time and provides open-source software to support future multi-Safe Haven projects. By working together as a network and with the support of local health boards the Safe Havens have successfully generated a reproducible methodology for aggregating and harmonising routine laboratory data that covered all major Scottish health boards for the benefit of future research projects wishing to use this important resource. We have identified and reported the heterogeneities (such as unit, reference range, etc) in data (structure, contents, test utilisation) and generated a harmonised, comparable cross-region health laboratory dataset for testing the process and methodology. The dataset created, being an important missing piece of puzzle of national level EHR data in Scotland, enables national level population health studies with biochemistry measurements, immunology etc.

The open-source software pipeline created can be utilised in harmonising Scottish laboratory datasets for future research projects and demonstrates that it is possible to bring together data from different laboratories to perform large scale analyses. The lessons learned can be reapplied to similar datasets in other regions of the UK and perhaps beyond. The codes developed can be adopted and transferred to incorporate knowledge on local data before they can be applied to the local ETL process.

Our work demonstrates that harmonisation of laboratory data on a national, cross-health board scale is possible but requires active input from all stakeholders, including laboratory staff, data teams, and clinicians, to ensure that decisions are made correctly, such as choosing the most sensible unit.

Through our study, we have observed and reported the difficulties that lack of data collection standard imposes on research utilising those data. The primary aim of healthcare data is to provide better care to the individual. The secondary usage of those

data enables the early diagnose, more accurate prognoses as well as personalised treatment, hence it can potentially improve the care provided by healthcare practitioners. Healthcare practitioners, researchers, and policymakers should recognise the importance of data collection standards and work together to reduce the data problem from the beginning at the collection.

4.1. Limitation and future work

There were several limitations of this study. First, missing data. Not all health boards in Scotland are covered by the regional Safe Haven Network and not all subtypes of laboratory test from health boards were covered. Regional Safe Havens are still working to access certain laboratory categories. Second, the population for this study is the SHARE cohort, which although aims to represent the wider Scottish population runs a risk from self-selection bias of people consenting to join. Despite that it should not affect the range of laboratory data available and impact from any bias should be minimal. Similarly, no inference or implication should be made regarding relative quality, accuracy or performance of any individual health board or region. Third, we have included 13 variables and 180 tests in our harmonisation pipeline which may not be sufficient for some research projects. For example, we mapped the sample collection date to "effectiveDate", however, the test executed date can be important since some tests e.g. potassium blood serum can be highly influenced by sample haemolysing (depending on the time gap between the sample been collected to the test been conducted). As this is an open-source pipeline researchers can adapt it to suit their needs and contribute enhancements. The focus here was on the health data landscape in Scotland. It may be possible that the pipeline can be adapted for other regionals/countries in the world. Fourth, the structure mapping was based on the semantics of each Safe Haven column and the definition of FHIR column [48]. We have not conducted an exhaustive examination of the contents of and standardising the terms used across Safe Havens.

Not all health boards were covered and not all tests were included. In this paper, we prioritised those columns/information considered essential by FHIR standard, as shown in Figure 2 and selected 180 most commonly used codes. Future work is needed to include missing health boards and tests. For those FHIR standard columns, future works are needed to standardise the contents for "specimenType", "category", "readCodeDescription" columns. In those columns, different terms and description can be used to describe the same thing. For example, "blood" and "serum" can both be used to describe the same sample collected for the same test in the "specimenType" column.

Those differences require knowledge and efforts on national language processing to harmonise them. Further work needed to standardise those three columns, as well as the other information that is reported in the local data but not included in the national dataset e.g. the “requested by” information. We are planning real-world studies to implement and refine the software.

4.2. Conclusion

Routinely collected laboratory data from patient samples is designed for clinical use and not for research. For statistical power and wider relevance research requires large datasets, however, routinely collected data has considerable heterogeneity and systematic differences existed in Scottish regional collected laboratory data. Here, we have developed open-source tools for researchers as well as regional Safe Haven staff to combine and harmonise Scottish laboratory data. We have demonstrated regional collected laboratory data can be harmonised as a national level, comparable dataset. Our work provided standard curation of laboratory data hence reduced duplicative efforts in research projects and aid reproducibility within and between regional studies in the future. It is our objective that this work will unlock potentially important new studies which were previously too onerous. Moreover, the work has shown health data curation as a complex, standalone, technical challenge of its own. Enhancive local data and laboratory knowledge is needed to harmonise the data. Enforcing data collection standards at the collection point can massively reduce the effects needed for the EHRs secondary usage.

Acknowledgement

This is research data for only a subset of health board data and no inference or implication should be made regarding relative quality, accuracy, or performance of individual health boards.

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Appendix

	link
Appendix A	https://github.com/HicResearch/ScottishLabData/blob/main/Paper%20Appendix/Appendix%20A.csv
Appendix B	https://github.com/HicResearch/ScottishLabData/blob/main/Paper%20Appendix/Appendix%20B.csv
Appendix C	https://github.com/HicResearch/ScottishLabData/blob/main/Paper%20Appendix/Appendix%20C.csv
Appendix D	https://github.com/HicResearch/ScottishLabData/blob/main/Paper%20Appendix/Appendix%20D.csv

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